

Chapter 5: Autoregressive Model

Autoregressive Model: Understand the stationarity and Markov property.

Autoregressive Model: Formulate the autoregressive (AR) model of order 1,

$$X_1 \sim f(x_1) \text{ and } X_t = \rho_1 X_{t-1} + \epsilon_t$$

where $\epsilon_t \sim N(0, \sigma^2)$ and $\theta = (\rho_1, \sigma^2)^T$, and describe how to find the MLE.

Deep Learning-based Approaches: Understand how Fully Visible Sigmoid Belief Network and Neural Autoregressive Density Estimation model the conditional probabilities and sequentially generate pixel values.

Chapter 6: Energy-based Model

Restricted Boltzmann Machine: Interpret the conditional independence between latent components given observed features:

$$f_{\theta}(\mathbf{z}|\mathbf{x}) = \prod_j f_{\theta}(z_j|\mathbf{x}).$$

Restricted Boltzmann Machine: Show that Gaussian distributions are energy-based models.

Restricted Boltzmann Machine: Understand how to train RBMs using Contrastive Learning.

Temporal Restricted Boltzmann Machine: Understand how Temporal RBMs model the temporal dependency.